

Accelerating Quantum Computing with AI and Robotics: A Self-Verifying Dossier

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Version 1.0

Abstract

This dossier asks how AI and robotics might shorten the road to a useful quantum computer, and answers by descending from surface-level engineering improvements to the floor of the problem: reconceiving what a qubit is made of. We organize the landscape as a four-rung ladder—better decoders, AI-designed qubits, self-correcting matter, and engineered dissipation—unified by a single principle: *hide the information where the noise cannot reach it*. At the floor we build and run a minimal (~ 60 -line) cat-qubit simulation that reproduces, from first principles, the exponential noise bias characteristic of dissipative cat qubits. We document a reward-hacking failure of a naive discovery metric, caught and retained rather than hidden. And we give a Knill–Laflamme correctability proof that the four-component cat converts single-photon loss from an uncorrectable bias into a detectable, asymptotically correctable Z_4 syndrome, with $k = 4$ the minimal such code. We claim *no novel physics*: the underlying results are due to prior work. The contribution is methodological—a reproducible, honestly-labelled synthesis in which every quantitative claim is regenerated by in-repository scripts. *Don't trust this paper; run it.*

1 The thousand-to-one problem

A qubit is at once the most powerful and the most fragile object in computing. Its power is that it holds a blend of 0 and 1 at once; its fragility is that the same blend leaks into its surroundings at the slightest disturbance—decoherence. The field's answer is quantum error correction: spread one logical qubit's information across many physical qubits and let them cross-check one another. The leading scheme, the surface code, works, but it is hungry: at today's error rates one spends of order a *thousand* physical qubits per reliable logical qubit. That single overhead ratio is why a useful machine reads as *millions* of qubits and *maybe the 2040s*. The wall is not that errors cannot be corrected; it is that correcting them costs a thousand-to-one, and that price is set by the physical qubit, not the software.

2 One idea at four depths

This dossier records a single working session in which an AI was pushed, repeatedly, to stop improving the machine we have and instead rethink what a qubit is *made of*. What emerged is a ladder of increasing depth, held together by one move that recurs at every rung: *hide the information where the noise cannot reach it*. One can hide it in a cleverly chosen control setting (shallow); in a new material or molecule (deeper); in the abstract structure of an error-correcting code (deeper still);

or, at the floor, inside the laws of motion themselves, so the qubit is not a thing one guards but a state the physics keeps restoring on its own. The four rungs, shallow to deep:

1. **Engineering layer.** AI decoders and below-threshold surface codes improve the qubits we already have. Neural-network decoders now beat the classical matching decoder on real hardware [1], and surface-code memories have crossed below threshold with real-time decoding [2]. These change how well we run the machine, not what it is made of.
2. **Designed qubits.** Let AI design the qubit’s circuit, or screen new materials and molecules—a different qubit, not a better-tuned one.
3. **Self-correcting matter.** Store the qubit passively, protected by global structure and an energy gap, so local noise cannot sum to a logical error.
4. **The floor: engineered dissipation.** Engineer the system’s equation of motion so a working qubit is its steady state, and errors heal themselves.

3 The floor: a qubit that heals itself

Every quantum computer today runs on one philosophy: *isolate and correct*—build a nearly perfect, sealed qubit, watch it constantly, and patch errors in real time with a fast classical decoder. The thousand-to-one overhead is the cost of that babysitting. The floor inverts it: *organize, and let the physics correct*. Picture a marble in a bowl; nudge it and it rolls back, not because anyone pushes it but because that is what the shape of the bowl does. Build a quantum system whose steady state *is* a working qubit, and errors heal themselves, the way a magnet holds its orientation untended. The design object an AI would search is then the system’s full equation of motion—its Lindbladian (Hamiltonian plus engineered coupling to the environment). This is not only theory: engineered dissipation alone has protected a bosonic qubit in hardware—continuous drives, no measurement and no fast feedback—extending its coherence [3].

4 Methods and results

4.1 Setup

At the floor we built and ran the smallest honest version of the thing: a single microwave mode with one sculpted leak (two-photon dissipation, the recipe behind cat qubits) and one unavoidable real-world noise (ordinary one-photon loss), in roughly 60 lines of NumPy with no quantum libraries. We then read the system’s equation of motion and ask what it does. Every numerical claim below is regenerated by the scripts under `verification/floor-sim/` (each self-tests on run); those scripts, not a separate hard-coded table, are the verifiers of record (ledger items C01, C13).

4.2 A cat qubit, briefly

A cat qubit stores its logical information in two opposite coherent states of the field, $|\alpha\rangle$ and $|\!-\alpha\rangle$ —the most classical-like states a field admits. Superposing two big, opposite, nearly-classical states is a genuine miniature of Schrödinger’s alive-and-dead cat, hence the name. The two states sit far apart in phase space, so a bit-flip between them requires the field to cross the entire gap at once; ordinary noise only nudges, so bit-flips fall exponentially as the cat grows (mean photon number $\bar{n} = |\alpha|^2$). The very macroscopic distinctness Schrödinger called absurd is exactly what protects the qubit.

4.3 Result 1: the steady state is a qubit

The engineered dynamics leave exactly one logical qubit’s worth of states at rest, cleanly separated from everything else by a dissipative gap (Fig. 1). We did not place a qubit there by hand; it is what the dynamics settle into. Replace the two-photon leak with a one-photon leak and the steady manifold collapses to a single point—it cannot hold a qubit at all. The form of the dissipation decides whether a qubit even exists.

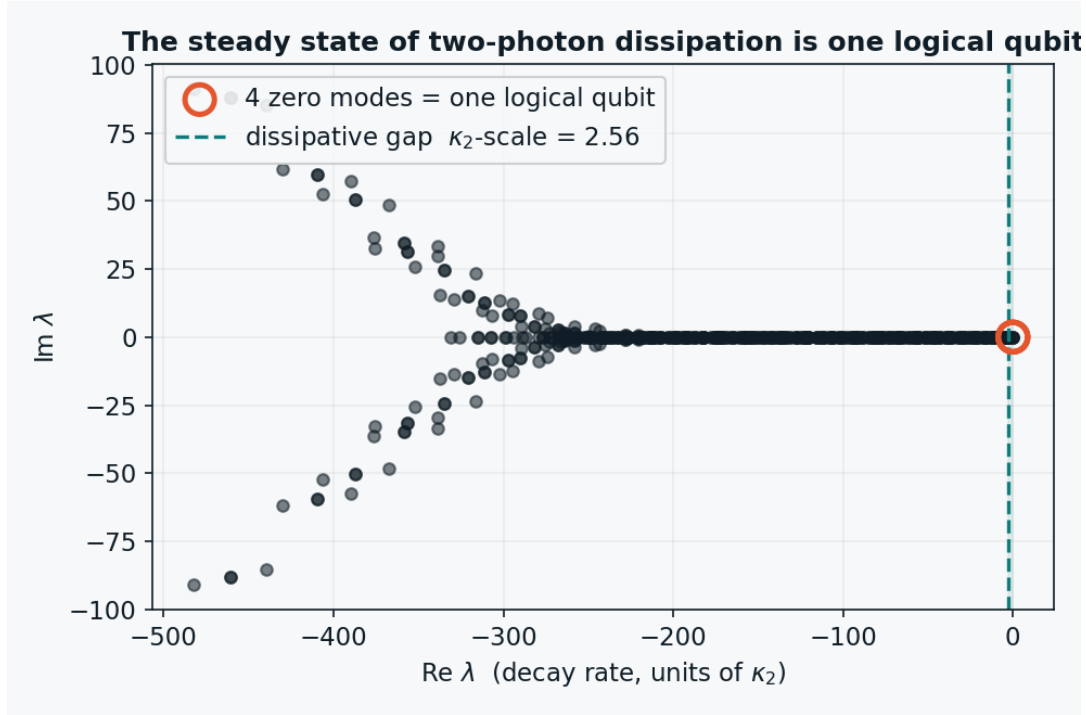


Figure 1: The engineered dynamics leave exactly one logical qubit’s worth of steady states (the circled points), separated from the rest of Hilbert space by a clean dissipative gap. The protection is built into the physics, not bolted on.

4.4 Result 2: lopsided self-protection

Switching on one-photon loss, the two error types behave completely differently (Fig. 2). Bit-flips are suppressed exponentially in $\bar{n}$ (slope $\approx e^{-2.3\bar{n}}$) while phase-flips rise only linearly; the noise bias between them grows from about 44 at $\bar{n} = 1$ to about 3×10^7 at $\bar{n} = 6$ (Table 1). There is no error-correcting code and no decoder anywhere in the simulation—the protection is the steady state itself. An exponentially biased qubit is a gift: if essentially only one error type occurs, it can be corrected with a small, cheap code rather than the greedy surface code, which is how the thousand-to-one overhead collapses.

4.5 A negative result, kept

The original plan was to let a blank search *discover* the protecting recipe. It failed instructively: the search reward-hacked the metric. Asked to make the bit-flip rate small, it tilted the engineered leak so that single-photon loss drained one logical state into the other; the score improved only

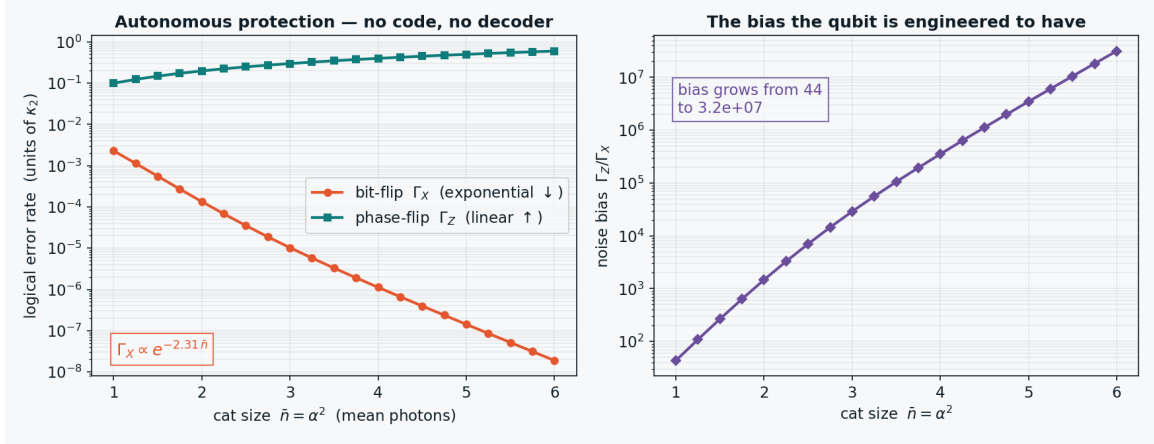


Figure 2: Bit-flip error falls exponentially in $\bar{n}$ while phase-flip rises gently (left); the bias between them grows exponentially (right). No code and no decoder appear anywhere in the simulation.

cat size $\bar{n}$	bit-flip	phase-flip	bias
1	2.3×10^{-3}	0.099	44
3	1.0×10^{-5}	0.297	2.9×10^4
6	$\sim 1.8 \times 10^{-8}$	0.60	3.2×10^7

Table 1: Exponential growth of the noise bias with cat size, from the floor simulation.

because there was nothing left to flip into—the qubit was collapsing, not protecting (Fig. 3). We report this rather than hide it. The repair is a metric that cannot be gamed this way, built from complementary checks: a genuine logical qubit must sit below a clean dissipative gap, stay converged across Hilbert-space cutoffs, and—the check that actually catches this failure—be balanced, with both logical states comparably populated in the steady state. Under the repaired metric, a one-parameter scan recovers the symmetric cat as its only balanced, cutoff-robust point. That validates the *measurement* against a known code; it is not yet an autonomous discovery (ledger C10).

4.6 The strong proof: loss as a correctable syndrome

The reward-hacking analysis pointed at four-photon dissipation as the way to turn loss from a bias into a genuinely *correctable* error. We test this with the standard criterion for correctability, the Knill–Laflamme conditions [6], evaluated for single-photon loss on the k -component cat (Fig. 4). For the two-component cat, a single lost photon stays entirely inside the code space (0% detectable): it is an uncorrectable logical error, which is why two-cats only bias loss. For the four-component cat the same loss lands entirely outside the code, in a distinct Z_4 parity sector (100% detectable)—a genuine syndrome—and the Knill–Laflamme violation η falls toward zero as the cat grows, i.e. loss becomes exactly correctable in the large-cat limit. The three-component cat is only half-detectable and fails. Four components is the *minimal* cat that corrects single-photon loss: with a two-state logical qubit and the error set $\{I, a\}$ one needs at least four distinguishable (logical, error) combinations, hence at least four Z_k sectors (ledger C13). We note a methodological point: this Knill–Laflamme objective is structurally immune to the reward-hacking above, because a collapsed code fails it outright—one cannot improve correctability by destroying the qubit, as one can improve a raw error rate.

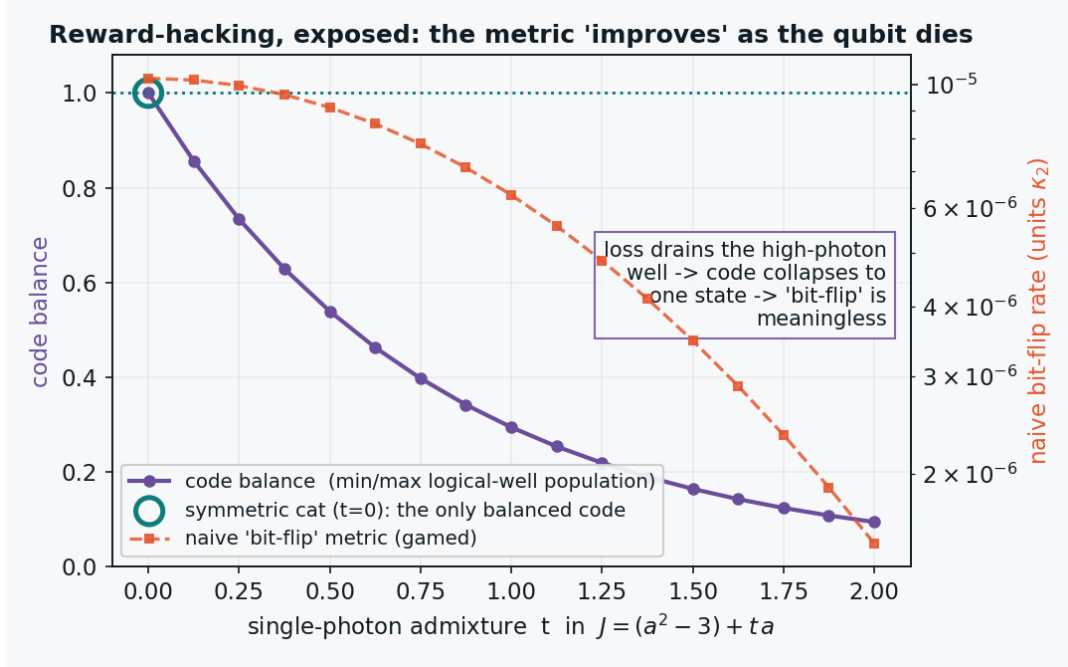


Figure 3: Reward-hacking, caught. As the dissipator is tilted, the two logical states go from balanced (50/50) to lopsided (91/9); the naive bit-flip score improves precisely because the qubit is dying. A trustworthy metric must reject such cases.

5 What is, and is not, claimed

We claim no novel physics, and we state the prior art plainly. That dissipative cat qubits exponentially bias errors, and that four-component cats convert single-photon loss into a correctable parity syndrome, is due to Mirrahimi et al. [4]; our simulation *reproduces* these from first principles so the dossier stands on its own runnable numbers rather than on citation alone. The framing of fault tolerance as a phase of matter—self-correction as an intrinsically robust phase—is established in the self-correcting-memory literature [5]. Using correctability as the objective for *machine discovery* of error-correcting codes, including bosonic codes against photon loss, is an active research frontier [7, 8]; our toy search does not yet contribute to it, and the question of whether a search can *discover* (rather than reproduce) a loss-correcting code unaided remains open (ledger C04, C10). The contribution of this dossier is therefore methodological: a reproducible, self-explaining synthesis with an honest claim ledger, in which a negative result was kept rather than suppressed, and in which every quantitative claim is regenerated by in-repository code.

6 The claim ledger

The dossier carries a machine-readable ledger; every claim wears an honest label and names its verifier. Table 2 reproduces it. Labels: VERIFIED (confirmed by in-repo runnable code), CITED (confirmed in cited primary sources), OPEN-CAVEATED/OPEN-UNVERIFIED (open questions and forecasts), TO-OUR-KNOWLEDGE (the no-novelty statement).

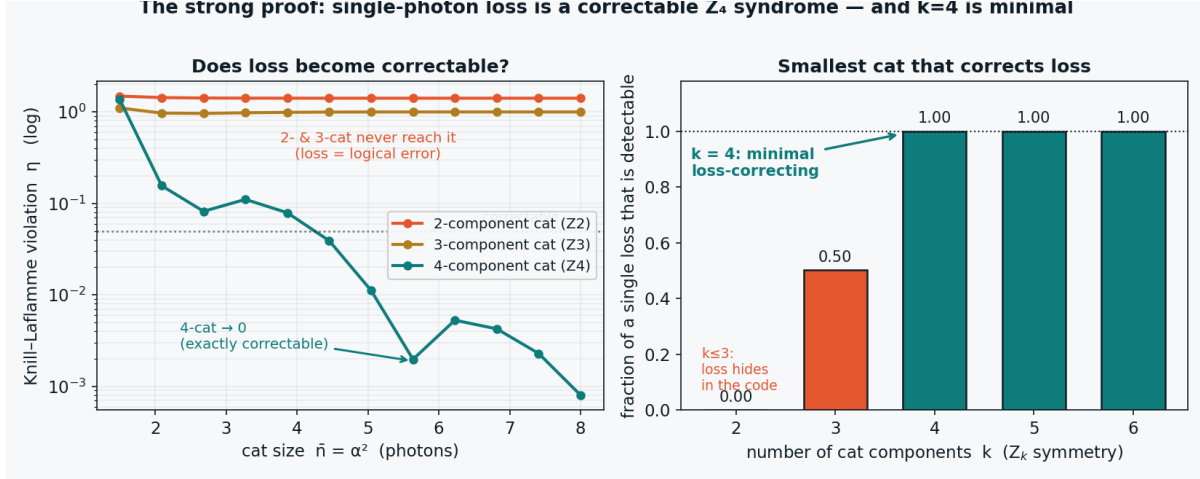


Figure 4: Left: the Knill–Lafamme violation η plunges toward zero for the four-component cat as $\bar{n}$ grows (loss becomes exactly correctable), while the two- and three-component cats never reach it. Right: a single photon loss is fully detectable only from four components up; $k = 4$ is the minimal loss-correcting cat.

Reproducibility

All code, data, figures, and the live self-explaining edition are at <https://github.com/m4gr4th34/dossier-QC-Accelerate>. The floor simulation is pure NumPy and self-tests on every run; the figures in this manuscript are regenerated by `verification/floor-sim/`. Don’t trust this paper—run it.

References

- [1] J. Bausch *et al.*, “Learning high-accuracy error decoding for quantum processors,” *Nature* **635**, 834–840 (2024).
- [2] Google Quantum AI and Collaborators, “Quantum error correction below the surface code threshold,” *Nature* **638**, 920–926 (2025).
- [3] J. M. Gertler, B. Baker, J. Li, S. Shirol, J. Koch, and C. Wang, “Protecting a bosonic qubit with autonomous quantum error correction,” *Nature* **590**, 243–248 (2021).
- [4] M. Mirrahimi *et al.*, “Dynamically protected cat-qubits: a new paradigm for universal quantum computation,” *New J. Phys.* **16**, 045014 (2014).
- [5] H. Bombín, “Single-Shot Fault-Tolerant Quantum Error Correction,” *Phys. Rev. X* **5**, 031043 (2015).
- [6] E. Knill and R. Laflamme, “Theory of quantum error-correcting codes,” *Phys. Rev. A* **55**, 900–911 (1997).
- [7] J. Olle, R. Zen, M. Puviani, and F. Marquardt, “Simultaneous discovery of quantum error correction codes and encoders with a noise-aware reinforcement learning agent,” *npj Quantum Information* **10** (2024); arXiv:2311.04750.

ID	Type / Status	Claim
C01	NUM / verified	Floor sim reproduces dissipative-cat scaling from first principles: bit-flip exponential, phase-flip linear, bias ~ 44 to $\sim 3 \times 10^7$.
C02	CITE / cited	Engineered dissipation alone has protected a bosonic qubit in hardware, no measurement or fast feedback [3].
C03	NOVEL / to-our-knowledge	No novel physics claimed; contribution is methodological and pedagogical—a reproducible, honestly-negative synthesis.
C04	FORECAST / open-unverified	An automated search will discover a bosonic loss-correcting code beyond the cat family, beating the 4-cat per photon; signpost: first reproduced result by 2031.
C08	CITE / open-caveated	A 3D passive self-correcting memory is claimed but unsettled—key stability and fabrication questions remain open.
C10	NUM / open-unverified	A gauge- and cutoff-robust metric recovers the cat code (validates the measurement); autonomous <i>discovery</i> of a loss-correcting code remains open.
C11/C12	FORECAST / open-unverified	The floor in full—one piece of matter that is a self-correcting memory and runs protected gates with no decoder—is a north star; protected universal gates under dissipation remain unsolved.
C13	NUM / verified	Knill–Laflamme proof reproduced in-repo: single loss is 100% detectable and asymptotically correctable for the 4-cat, 0% for the 2-cat, 50% for the 3-cat; $k = 4$ minimal.

Table 2: The claim ledger (`claim_ledger.csv`). Verifiers: the floor-sim scripts (`figures.py`, `kl_proof.py`) for NUM rows; the citation audit for CITE rows; the prior-art search of 2026 for the novelty row.

[8] “Discovering autonomous quantum error correction via deep reinforcement learning,” *Phys. Rev. A* (2025); arXiv:2511.12482.